

易宇豪 (Yuhao Yi)

特聘研究员 (APPOINTED RESEARCHER)

华西医院 (West China Hospital),

四川大学 (Sichuan University).

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研究领域 (Areas of Specialization)

数字病理 (Digital Pathology) • 机器学习 (Machine Learning) • 优化算法 (Optimization Algorithms)

工作经历 (Professional Preparation)

- | | |
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| 2024.08-至今 | 特聘研究员
四川大学华西医院
临床病理研究所
研究方向: 病理人工智能、医学图像处理、生物信息学 |
| 2022.08-2024.07 | 特聘研究员
四川大学计算机学院
数据智能与计算艺术实验室
研究方向: 分布式机器学习、优化、近似算法 |
| 2020.08-2022.07 | 博士后 (Postdoctoral Researcher)
Division of Decision and Control Systems (DCS), EECS,
KTH Royal Institute of Technology, Sweden.
Supervisor: Karl H. Johansson.
研究方向: (随机) 传播过程控制与优化、网络系统鲁棒性 |
| 2019.03-2020.05 | 博士后 (Postdoctoral Researcher)
Networked Systems Lab, Department of Computer Science,
Rensselaer Polytechnic Institute, U.S.A.
Supervisor: Stacy Patterson.
研究方向: 网络化控制系统、扩散过程 |

教育经历 (Education)

- 2019.01 理学博士，计算机软件与理论
上海市智能信息处理重点实验室
计算机学院，复旦大学
导师：章忠志
研究方向：随机算法、谱图理论
- 2013.07 理学学士，计算机科学与技术，复旦大学

论文 (Selected Publications)

会议 (CONFERENCES)

- 11 Wenchuan Zhang, Penghao Zhang, Jingru Guo, Jie Chen, Shuwan Zhang, Zhang Zhang, **Yuhao Yi**^{*}, Hong Bu, *Patho-R1: A Multimodal Reinforcement Learning-Based Pathology Expert Reasoner*. AAAI-2026. (中国计算机学会推荐 A 类会议, 人工智能)
- 10 Wenchuan Zhang, Jingru Guo, Hengzhe Zhang, Penghao Zhang, Jie Chen, Shuwan Zhang, Zhang Zhang, **Yuhao Yi**^{*}, Hong Bu, *Patho-AgentRAG: Towards Multimodal Agentic Retrieval-Augmented Generation for Pathology VLMs via Reinforcement Learning*. AAAI-2026. (中国计算机学会推荐 A 类会议, 人工智能)
- 9 Hong Liu, Liren Shan, Han Bao, Ronghui You^{*}, **Yuhao Yi**^{*}, Jiancheng Lv, *LiD-FL: Towards List Decodable Federated Learning*. AAAI-2025. (中国计算机学会推荐 A 类会议, 人工智能)
- 8 **Yuhao Yi**, Liren Shan, Philip E. Paré, Karl H. Johansson, *Dynamic Curing and Network Design in SIS Epidemic Processes*. CDC-2024. (控制三大会议)
- 7 **Yuhao Yi**, Ronghui You, Hong Liu, Changxin Liu, Yuan Wang^{*}, and Jiancheng Lv^{*}, *Near-Optimal Resilient Aggregation Rules for Distributed Learning Using 1-Center and 1-Mean Clustering with Outliers*. AAAI-2024. (中国计算机学会推荐 A 类会议, 人工智能)
- 6 **Yuhao Yi**, Yuang Wang, Xingkang He, Stacy Patterson, and Karl H. Johansson, *A Sample-Based Algorithm for Approximately Testing r -Robustness of a Digraph*. CDC-2022: 6478-6483. (控制三大会议)
- 5 **Yuhao Yi** and Stacy Patterson, *Disagreement and Polarization in Two-Party Social Networks*. (IFAC World Congress 2020) IFAC-PapersOnLine, 53(2), 2568-2575. (控制三大会议)
- 4 (字典序) Huan Li, Richard Peng, Liren Shan, **Yuhao Yi**, and Zhongzhi Zhang, *Current Flow Group Closeness Centrality for Complex Networks*. WWW-2019: 961-971. (中国计算机学会推荐 A 类会议, 交叉/综合/新兴)
- 3 **Yuhao Yi**, Liren Shan, Huan Li, and Zhongzhi Zhang^{*}, *Biharmonic Distance Related Centrality for Edges in Weighted Networks*. IJCAI-ECAI-2018: 3620-3626. (中国计算机学会推荐 A 类会议, 人工智能)

- 2 Liren Shan, **Yuhao Yi**, Zhongzhi Zhang*, *Improving Information Centrality of a Node in Complex Networks by Adding Edges*. IJCAI-ECAI-2018: 3535-3541. (中国计算机学会推荐 A 类会议, 人工智能)
- 1 **Yuhao Yi**, Bingjia Yang, Zhongzhi Zhang*, and Stacy Patterson*, *Bi-harmonic Distance and the Performance of Second-Order Consensus Networks with Stochastic Disturbances*. ACC-2018: 4943-4950. (控制三大会议)

期刊 (JOURNAL ARTICLES)

- 13 Changxin Liu*, Yanghao Li, **Yuhao Yi***, Karl H. Johansson. *Byzantine-Robust and Communication-Efficient Distributed Learning via Compressed Momentum Filtering*. IEEE Trans. Neural Networks Learn. Syst. 2026. (中科院 1 区, 计算机)
- 12 Xunxi Lu, Bin Luo, Yani Wei, Wenchuan Zhang, Xiaoyan Wu, Jie Chen, Huijuan Shi, Jingping Yuan, Hong Bu, Yuhao Yi, Zongchao Guo. *Integration of histopathological characteristics by machine learning improves the prediction of neoadjuvant immunotherapy response in triple-negative breast cancer*. J. Pathol. (中科院 1 区, 医学)
- 11 Fengling Li, Yani Wei, Wenchuan Zhang, Yuanyuan Zhao, Jing Fu, Xiuli Xiao, Yan Qiu, **Yuhao Yi***, Yongquan Yang*, and Hong Bu. *AI-powered prediction model for neoadjuvant chemotherapy efficacy: comprehensive analysis of breast cancer histological images*. NPJ Precis. Oncol. 2025, 9, no.1, pp. 239. (中科院 1 区, 医学)
- 10 **Yuhao Yi***, Liren Shan, Shijie Wang, Philip E. Paré, Karl H. Johansson, *Efficient Algorithms for Dynamic Curing and Network Design in SIS Epidemic Processes*. IEEE Trans. Netw. Sci. Eng., 2025, 13, 1472-1484. (中科院 2 区, 计算机)
- 9 Xiaoyan Wu, Yiman Li, Jilong Chen, Jie Chen, Wenchuan Zhang, Xunxi Lu, Xiaorong Zhong, Min Zhu, **Yuhao Yi***, Hong Bu, *Multimodal recurrence risk prediction model for HR+/HER2-early breast cancer following adjuvant chemo-endocrine therapy: integrating pathology image and clinical pathological features*. Breast Cancer Res., 2025, 27(1), pp. 27. (中科院 2 区, 医学)
- 8 **Yuhao Yi***, Liren Shan, Philip E. Paré, and Karl H. Johansson, *Edge Deletion Algorithms for Minimizing Spread in SIR Epidemic Models*. SIAM J. Control Optim., 2022, 60(2), S246-S273. (中国自动化学会推荐 A 类期刊)
- 7 **Yuhao Yi**, Bingjia Yang, Zuobai Zhang, Zhongzhi Zhang*, and Stacy Patterson*, *Bi-harmonic Distance-Based Performance Metric for Second-Order Noisy Consensus Networks*. IEEE Trans. Inf. Theory, 2022, 68(2), 1220-1236. (中国计算机学会推荐 A 类期刊, 计算机理论)
- 6 **Yuhao Yi***, Anirban Das, Bassam Bamieh, Zhongzhi Zhang, and Stacy Patterson, *Diffusion and consensus in a weakly coupled network of networks*. IEEE Trans. Control Netw. Syst., 2021, 8(4), 1601-1612. (中国自动化学会推荐 A 类期刊)
- 5 **Yuhao Yi***, Timothy Castiglia, and Stacy Patterson, *Shifting Opinions in a Social Network through Leader Selection*. IEEE Trans. Control Netw. Syst., 2021, 8(3), 1116-1127. (中国自动化学会推荐 A 类期刊)

- 4 (字典序) Huan Li, Stacy Patterson, **Yuhao Yi**, and Zhongzhi Zhang, *Maximizing the Number of Spanning Trees in a Connected Graph*. IEEE Trans. Inf. Theory, 2020, 66(2), 1248-1260. (中国计算机学会推荐 A 类期刊, 计算机理论)
- 3 **Yuhao Yi**, Zhongzhi Zhang*, and Stacy Patterson, *Scale-Free Loopy Structure is Resistant to Noise in Consensus Dynamics in Complex Networks*. IEEE Trans. Cybern., 2020, 50(1), 190-200. (中国自动化学会推荐 A 类期刊, 计算机类川大 B)
- 2 Stacy Patterson*, **Yuhao Yi**, and Zhongzhi Zhang. *A Resistance Distance-Based Approach for Optimal Leader Selection in Noisy Consensus Networks*. IEEE Trans. Control Netw. Syst., 2019, 6(1): 191-201. (中国自动化学会推荐 A 类期刊)
- 1 **Yuhao Yi**, Zhongzhi Zhang*, Liren Shan, and Guanrong Chen. *Robustness of First- and Second-Order Consensus Algorithms for a Noisy Scale-Free Small-World Koch Network*. IEEE Trans. Control Syst. Technol., 2017, 25 (1), 342-350. (中国自动化学会推荐 A 类期刊)

讲座 (Talks)

- 2025.03 把模型关进笼子——事实校正的后训练和后验方法. 西湖大学, 杭州.
- 2019.11 *Fast Approximation Algorithms and Complexity Analysis for Design of Networked Systems*. Algorithms & Randomness Center, Georgia Institute of Technology. Atlanta, GA, USA.
- 2018.01 *Laplacian-based Distance Analysis of First- and Second-Order Network Coherence*. International School and Conference on Network Science (NetSci-X 2018). Hangzhou, China.

审稿人 (Reviewer for)

Conferences: ICML; AAAI; CVPR; ECCV; AISTATS; ACC; CDC.
Journals: TAC; Automatica; SICON; TKDE; TCNS, TNSE; ACM Health.

项目 (Project)

- 2026-2027 具有统计保证的可信机器学习在病理人工智能中的应用
四川省自然科学基金, 青年科学基金 (B 类), 负责人
- 2025 数字病理图像处理软件
四川省首批创新医疗器械融合应用试点项目 (四川省经信厅), 负责人
- 2024-2026 弹性网络化控制系统中的信息过滤与冗余检验算法
国家自然科学基金委, 青年科学基金 (C 类), 负责人
- 2020-2022 HiSS: Humanizing the Sustainable Smart City, Digital Futures (Sweden), 参与
- 2016-2020 A Machine Learning Framework for the Internet of Things, NSF (USA), 参与
- 2018-2019 复杂网络中的偏好游走及其应用, NSFC 面上项目, 参与 (排名第 4)
- 2018-2019 并行化和空间高效的参数算法, NSFC 面上项目, 参与

教学 (Teaching)

- 2023 软件学院，四川大学，数据结构与算法（留学生课程，全英文）
2023-2025 计算机学院，四川大学，操作系统原理、课程设计（本科专业核心课程）

奖项 (Awards)

- 2023 入选四川省天府峨眉计划
2021 University Dissertation Award 2020, Fudan University (校级优博论文，计算机科学技术学院 1 人)
2020 Wilkes Award, Best paper published in the Computer Journal in 2019 (CCF-B 类期刊年度最佳论文)